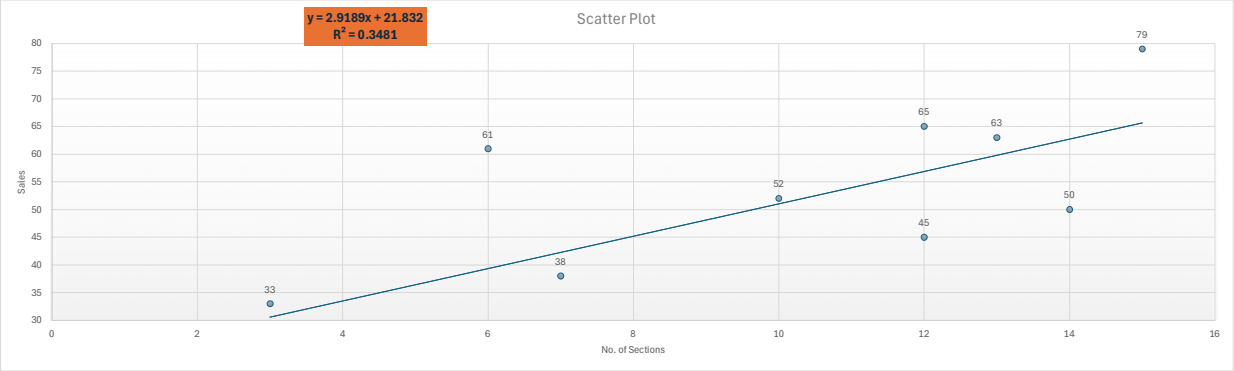
Q1. Campus Stores has been selling the *Believe It or Not: Wonders of Statistics Study Guide* for 12 semesters and would like to estimate the relationship between sales and number of sections of elementary stati stics taught in each semester. The following data have been collected:

a) Develop the estimating equation that best fits the data. (**10 marks**)
b) Calculate the sample coefficient of determination and the sample coefficient of correlation. (**5 marks**)

No. of Sections	Sales (units)	Correlation (r)
3	33	0.589993383
7	38	=CORREL(B11:B22, A11:A22)
6	24	
6	61	
10	52	
12	45	
12	65	
13	82	
12	29	
13	63	
14	50	
15	79	



Q2: A community health organisation is studying the average number of pediatric care hours children stay in the local clinic. A random sample of 50 children who received care in the clinic over the past year is taken. The number of care hours each child stayed is recorded (rounded to the nearest hour).

a. Find the mean and standard deviation of the time spent in the clinic. (**5 marks**)
b. Develop a 95% confidence interval for the average time spent in the clinic. (**5 marks**)
c. Experience has shown that the minimum required time for effective pediatric care is 37 hours. Test this claim at the 5% significance level using the data above. (**10 marks**)

Values	Mean	Standard Deviation	95% Confidence Interval	Sample Size	SE	Hypothesized Mean	t-statistic	t_critical (One-tailed, α = 0.05)
41	38.4	2.294625486	0.652125348	50	0.324509048	37	4.314209441	1.676550893
40							**t-stat > t_critical, reject H_0.**	
39	=AVERAGE(A43:A92)	=STDEV.S(A43:A92)	=CONFIDENCE.T(0.05, STDEV.S(A43:A92), 50)	=COUNT(A43:A92)	=C43/SQRT(E43)		=(B43-G43)/F43	=T.INV(0.95, E43-1)
38								
36								
37								
40								
41								
35								
42								
38								
37								
39								
40								
41								
34								
36								
37								
39								
38								
36								
38								
35								
40								
42								
36								
41								
43								
40								
39								
37								
38								
39								
38								
41								
37								
35								
36								
42								
38								
39								
37								
40								
36								
38								
36								
34								
38								
41								
42								

DATA:
Sample mean $\bar{x} = 38.4$
Hypothesized mean $\mu_0 = 37$
Standard deviation $s = 2.2946$
Sample size $n = 50$
Significance level $\alpha = 0.05$

STATE HYPOTHESES:
Null Hypothesis (H_0): $\mu = 37$
Alternative Hypothesis (H_1): $\mu > 37$ (since we're testing if the average time is **more than** the minimum) right-tailed test.

TEST STATISTIC:
4.314

CRITICAL VALUE:
Degrees of freedom: df = 49
tcritical = T.INV(0.95, 49) ≈ 1.676

DECISION:
At the 5% significance level, we conclude that the **average time spent is significantly greater than 37 hours**, supporting the claim that pediatric care exceeds the minimum required time.

Q3. Picosoft Ltd., a supplier of operating system software for personal computers, was planning the initial public offering of its stock in order to raise sufficient working capital to finance the development of a radically new, seventh-generation integrated system. With current earnings of $1.61 a share, Picosoft and is its underwriters were contemplating an offering price of $21, or about 13 times earnings. In order to check the appropriateness of this price, they randomly chose seven publicly traded software firms and found that their average price/ earnings ratio was 11.6, and the sample standard deviation was 1.3. At α =0.02, can Picosoft conclude that the stocks of publicly traded software firms have a P/E ratio that is significantly different from 13? (**10 marks**)

GIVEN:

population mean (μ_0) = 13, sample mean (x) = 11.6, s = 1.3, n = 7, (α) = 0.02, (df) = n - 1 = 6

Test: Is 13 significantly different from sample?

T-STATISTIC	CRITICAL T-VALUE
-2.849270643	3.142668403
=(11.6 - 13) / (1.3 / SQRT(7))	=T.INV.2T(0.02, 6)

If $|t| > t_{\alpha/2}$, **reject** the null hypothesis.
Here: | −2.849| < 3.143 → **Do not reject** the null hypothesis.
At the 0.02 significance level, the calculated t-statistic (−2.85) does **not exceed** the critical t-value (±3.143).
Therefore, we **fail to reject the null hypothesis**.

CONCLUSION:
There is **not enough statistical evidence** to conclude that the average P/E ratio of publicly traded software firms is significantly different from 13. Thus, based on this sample, Picosoft **cannot conclude** that 13 is an inappropriate benchmark P/E ratio. The planned offering price based on 13 times earnings **may be considered reasonable** within the bounds of

Q4. According to the U.S. Census Bureau, the average American generates 4.4 pounds of garbage per day.
Suppose we believe that because of recycling and a greater emphasis on the environment, the figure is now lower. To test this notion, we take a random sample of Americans and have them keep a log of their garbage for a day. We record and analyse the results by using a statistical computer package. The output follows. Describe the sample. What statistical decisions can be made on the basis of this analysis? Let alpha be .05. Assume that pounds of garbage per day are normally distributed in the population. Discuss any substantive results.

One-Sample Z

```
Test of mu = 4.4 vs < 4.4
The assumed standard deviation = 0.866
                          95%
                        Upper
N   Mean  SE Mean      Bound      Z      P
22  3.969    0.185      4.273  -2.33  0.010
```

Sample Description:
From the output:
Sample size (N): 22
Sample mean: 3.969 pounds
Standard Error (SE Mean): 0.185
Assumed population standard deviation: 0.866
Upper 95% confidence bound: 4.273
Z value: -2.33
P-value: 0.010

Hypothesis Test:
We are testing:
Null Hypothesis (H_0): μ = 4.4
Alternative Hypothesis (H_1): μ < 4.4
This is a left-tailed one-sample Z-test.

Decision Rule:
Significance level (α) = 0.05
The p-value = 0.010
Since p-value < α (0.010 < 0.05), we reject the null hypothesis.

Conclusion:
There is **sufficient statistical evidence** at the 0.05 significance level to conclude that the **mean amount of garbage generated per person per day is less than 4.4 pounds**. This supports the belief that recycling and environmental efforts may be having a positive impact.

This result suggests that Americans, on average, are now generating **significantly less waste** than the previous average of 4.4 pounds per day. This could be due to increased recycling efforts, better environmental awareness, or changes in consumption behavior.